Parametric Inference

Worked Solutions to Homeworks 1–3

Companion to `PI_Exam_Revision.pdf`

Read these only after a cold attempt. Each solution states the key idea first, so you can take the hint and go back to the problem before reading the algebra.

Contents

1 Homework 1 — Unbiased estimation and UMVUE

Q1. X has pmf $f(x \mid \theta) = \dfrac{\Gamma(\alpha+x)}{\Gamma(\alpha)\,\Gamma(x+1)}\,\theta^\alpha(1-\theta)^x$, $x = 0, 1, 2, \dots$, with $0 < \theta < 1$ unknown and $\alpha > 0$ known. For any given α, does a UMVUE of θ exist? Find it when it does.

Key idea

Substitute $p = 1-\theta$ and use the negative binomial series $(1-p)^{-\alpha} = \sum_{k\ge0} c_k p^k$ with $c_k = \Gamma(\alpha+k)/(\Gamma(\alpha)k!)$. Matching power-series coefficients gives the unique unbiased estimator, and uniqueness makes it the UMVUE.

Solution. Write $c_x = \dfrac{\Gamma(\alpha+x)}{\Gamma(\alpha)\,x!}$, so $f(x \mid \theta) = c_x\theta^\alpha(1-\theta)^x$. Note first that this is a one-parameter exponential family: $f(x \mid \theta) = \theta^\alpha c_x \exp\{x\log(1-\theta)\}$ with $T(x) = x$ and natural parameter $\eta = \log(1-\theta)$ ranging over $(-\infty, 0)$, an open interval. Hence X **is complete and sufficient**, and by Lehmann–Scheffé any unbiased estimator (necessarily a function of X) is *the* UMVUE.

It remains to find one. Put $p = 1-\theta$. Unbiasedness of T reads

$$\sum_{x\ge0} T(x)\,c_x\,(1-p)^\alpha p^x = 1-p \quad\Longleftrightarrow\quad \sum_{x\ge0} T(x)c_x p^x = (1-p)^{1-\alpha} \qquad \forall\, p\in(0,1).$$

Now use the negative binomial series $(1-p)^{-\alpha} = \sum_{k\ge0} c_k p^k$, so

$$(1-p)^{1-\alpha} = (1-p)\sum_{k\ge0} c_k p^k = \sum_{k\ge0} c_k p^k - \sum_{k\ge0} c_k p^{k+1} = \sum_{x\ge0}(c_x - c_{x-1})p^x, \qquad c_{-1} := 0.$$

Two power series agree on an interval iff their coefficients agree, so $T(x)c_x = c_x - c_{x-1}$, i.e.

$$T(x) = 1 - \frac{c_{x-1}}{c_x} = 1 - \frac{\Gamma(\alpha+x-1)}{\Gamma(\alpha)(x-1)!}\cdot\frac{\Gamma(\alpha)\,x!}{\Gamma(\alpha+x)} = 1 - \frac{x}{\alpha+x-1},$$

using $\Gamma(\alpha+x) = (\alpha+x-1)\Gamma(\alpha+x-1)$ and $x! = x\,(x-1)!$. Therefore

$$\boxed{T(X) = \frac{\alpha-1}{\alpha+X-1} \quad (X\ge1), \qquad T(0) = 1.}$$

For $\alpha \ne 1$ the single formula $(\alpha-1)/(\alpha+X-1)$ covers $X = 0$ as well. For $\alpha = 1$ (the geometric case) all $c_x = 1$ and the estimator is $T(X) = \mathbb{I}\{X = 0\}$.

Existence. The coefficient matching determines T uniquely, so an unbiased estimator exists for **every** $\alpha > 0$. Its variance is finite because T is bounded: $|T(x)| \le \max\{1, |\alpha-1|/\alpha\}$. Hence the UMVUE exists for every $\alpha > 0$ and is the estimator boxed above.

Check ($\alpha = 2$). Then $c_x = x+1$ and $T(x) = 1/(x+1)$, so $\mathbb{E}_\theta T = \sum_{x\ge0}\frac{1}{x+1}(x+1)\theta^2(1-\theta)^x = \theta^2\cdot\frac{1}{\theta} = \theta$. ✓

Q2. $X \sim \text{Poisson}(\lambda)$, $\lambda > 0$. Find the UMVUE of $e^{c\lambda}$ for a given $-\infty < c < \infty$.

Key idea

Poisson is a complete-sufficient exponential family, so *any* unbiased estimator is the UMVUE. Write unbiasedness as a power-series identity and match coefficients of λ^x.

Solution. X is complete sufficient ($T(x) = x$, $c(\lambda) = \log\lambda$, natural space $\mathbb{R}$). Let T be unbiased:

$$\sum_{x\ge0} T(x)\,\frac{e^{-\lambda}\lambda^x}{x!} = e^{c\lambda} \iff \sum_{x\ge0}\frac{T(x)}{x!}\lambda^x = e^{(c+1)\lambda} = \sum_{x\ge0}\frac{(c+1)^x}{x!}\lambda^x \qquad \forall\lambda>0.$$

Both sides are power series in λ converging on $(0,\infty)$, so the coefficients match: $T(x) = (1+c)^x$. Hence

$$\boxed{T(X) = (1+c)^X}$$

is the unique unbiased estimator, and therefore the UMVUE by Lehmann–Scheffé.

The variance is finite: $\mathbb{E}_\lambda T^2 = \sum_x (1+c)^{2x} e^{-\lambda}\lambda^x/x! = \exp\{\lambda[(1+c)^2-1]\} < \infty$, so $\operatorname{Var}_\lambda T = e^{\lambda[(1+c)^2-1]} - e^{2c\lambda}$.

Watch out

With n iid observations, $S = \sum X_i \sim \mathrm{Poi}(n\lambda)$ is complete sufficient and the same argument gives $T = (1+c/n)^S$ as the UMVUE of $e^{c\lambda}$. Note the estimator can be negative when $c < -1$, or oscillate in sign when $c < -2$ — a standard illustration that a UMVUE need not be sensible.

Q3. $X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$, $\theta \in \mathbb{R}$. Find the UMVUE of $\mathbb{P}\{X_1 > 0\}$.

Key idea

Route B: Rao–Blackwellise the crude unbiased estimator $\mathbb{I}\{X_1 > 0\}$ over the complete sufficient statistic $\overline{X}$. The only work is the conditional law of X_1 given $\overline{X}$.

Solution. Here $\psi(\theta) = \mathbb{P}_\theta(X_1 > 0) = \Phi(\theta)$, where Φ is the standard normal cdf. $\overline{X}$ is complete sufficient ($N(\theta,1)$ is an exponential family with natural parameter θ over $\mathbb{R}$). Start from the unbiased estimator $T_0 = \mathbb{I}\{X_1 > 0\}$, which satisfies $\mathbb{E}_\theta T_0 = \Phi(\theta)$, and Rao–Blackwellise:

$$T^* = \mathbb{E}\big[\mathbb{I}\{X_1 > 0\} \bigm| \overline{X} = t\big] = \mathbb{P}\big(X_1 > 0 \mid \overline{X} = t\big).$$

Conditional law. $(X_1, \overline{X})$ is jointly normal with $\operatorname{Cov}(X_1, \overline{X}) = \frac{1}{n}\operatorname{Var}(X_1) = \frac{1}{n} = \operatorname{Var}(\overline{X})$. Hence

$$\mathbb{E}[X_1 \mid \overline{X} = t] = \theta + \frac{\operatorname{Cov}(X_1, \overline{X})}{\operatorname{Var}(\overline{X})}(t - \theta) = t, \qquad \operatorname{Var}(X_1 \mid \overline{X}) = \operatorname{Var}(X_1) - \frac{\operatorname{Cov}^2}{\operatorname{Var}(\overline{X})} = 1 - \frac{1}{n} = \frac{n-1}{n}.$$

(Equivalently: $X_1 - \overline{X} \perp\!\!\!\perp \overline{X}$, and $\operatorname{Var}(X_1 - \overline{X}) = 1 - 1/n$.) So $X_1 \mid \overline{X} = t \;\sim\; N\big(t, \frac{n-1}{n}\big)$ and

$$T^* = \mathbb{P}\Big(N\big(t, \tfrac{n-1}{n}\big) > 0\Big) = \Phi\left(\frac{t}{\sqrt{(n-1)/n}}\right).$$

Since T^* is unbiased and a function of the complete sufficient statistic, Lehmann–Scheffé gives

$$\boxed{\text{UMVUE of } \mathbb{P}\{X_1 > 0\} \;=\; \Phi\Big(\sqrt{\tfrac{n}{n-1}}\;\overline{X}\Big).}$$

Sanity check: as $n \to \infty$ the factor $\to 1$ and $T^* \to \Phi(\theta)$ by the LLN and continuity. ✓

Q4. For a sequence $\{a_x\}$ of positive constants let $A(\theta) = \sum_{x \ge 0} a_x \theta^x < \infty$ for $0 < \theta < b$, and let X have pmf $a_x\theta^x/A(\theta)$. Does the UMVUE of $\mathbb{E}(X)$ attain the Cramér–Rao lower bound? Verify.

Key idea

Yes. In a power-series family the score is exactly $\frac{1}{\theta}\{X - \mathbb{E}X\}$, which is the equality condition in the CRLB. The verification also gives the identity $\operatorname{Var}_\theta X = \theta\,\psi'(\theta)$.

Solution. Write $\psi(\theta) = \mathbb{E}_\theta X$. This is a one-parameter exponential family ($T(x) = x$, $c(\theta) = \log\theta$, natural space $(-\infty, \log b)$, an open interval), so X is complete sufficient; X is unbiased for $\psi(\theta)$; hence X **is the UMVUE of** $\mathbb{E}(X)$.

Step 1: the mean. Using the operator $\theta\frac{d}{d\theta}$, which turns $\sum a_x\theta^x$ into $\sum x a_x\theta^x$,

$$\psi(\theta) = \mathbb{E}_\theta X = \frac{1}{A(\theta)}\sum_x x\,a_x\theta^x = \frac{\theta A'(\theta)}{A(\theta)} = \theta\,L'(\theta), \qquad L := \log A.$$

Step 2: the variance. Applying the operator twice, $\sum_x x^2 a_x\theta^x = \theta\frac{d}{d\theta}\big(\theta A'\big) = \theta A' + \theta^2 A''$, so

$$\operatorname{Var}_\theta X = \frac{\theta A' + \theta^2 A''}{A} - \Big(\frac{\theta A'}{A}\Big)^2 = \theta L' + \theta^2\Big(\frac{A''}{A} - \frac{(A')^2}{A^2}\Big) = \theta L' + \theta^2 L''.$$

On the other hand $\psi' = (\theta L')' = L' + \theta L''$, so $\theta\psi' = \theta L' + \theta^2 L''$. Hence

$$\boxed{\operatorname{Var}_\theta X = \theta\,\psi'(\theta).}$$

Step 3: Fisher information. $\log f(x \mid \theta) = \log a_x + x\log\theta - \log A(\theta)$, so

$$\frac{\partial}{\partial\theta}\log f = \frac{x}{\theta} - \frac{A'(\theta)}{A(\theta)} = \frac{1}{\theta}\Big(x - \frac{\theta A'}{A}\Big) = \frac{1}{\theta}\big[x - \psi(\theta)\big].$$

Therefore $I(\theta) = \operatorname{Var}_\theta\big(\frac{1}{\theta}(X-\psi)\big) = \dfrac{\operatorname{Var}_\theta X}{\theta^2} = \dfrac{\psi'(\theta)}{\theta}$.

Step 4: compare. The CRLB for estimating $\psi(\theta)$ from one observation is

$$\frac{[\psi'(\theta)]^2}{I(\theta)} = \frac{[\psi'(\theta)]^2\,\theta}{\psi'(\theta)} = \theta\,\psi'(\theta) = \operatorname{Var}_\theta(X).$$

So the UMVUE X **attains the CRLB exactly, for every** $\theta \in (0,b)$. □

Two remarks worth a line in an exam script. (i) The structural reason is visible in Step 3: the score has the form $k(\theta)[T(x) - \psi(\theta)]$ with $k(\theta) = 1/\theta$, which is precisely the equality condition in the Cauchy–Schwarz proof of the CRLB. (ii) With n iid observations, $\overline{X}$ is the UMVUE, $\operatorname{Var}(\overline{X}) = \theta\psi'(\theta)/n$, and the bound $[\psi']^2/(nI)$ matches again.

Q5. $Y_i \sim N(\beta x_i, \sigma^2)$ independent, $i = 1,\dots,n$, with $\beta\in\mathbb{R}$ and $\sigma^2\in(0,\infty)$ unknown and $x_1,\dots,x_n$ known non-zero constants. Show the least squares estimate of β and the associated residual variance are UMVUEs. Are they independent?

Key idea

Recognise a two-parameter exponential family with $S = (\sum Y_i^2,\ \sum x_iY_i)$, whose natural parameter space is the open half-plane $\mathbb{R}\times(-\infty,0)$ — so S is complete sufficient. Both estimators are functions of S and unbiased. For independence, use orthogonality of the fitted and residual parts (or Basu).

Solution. *Complete sufficiency.* The joint density is

$$f(\mathbf{y}) = (2\pi\sigma^2)^{-n/2}\exp\Big\{-\frac{1}{2\sigma^2}\sum_i(y_i-\beta x_i)^2\Big\} = \underbrace{(2\pi\sigma^2)^{-n/2}e^{-\beta^2\sum x_i^2/2\sigma^2}}_{p(\beta,\sigma^2)}\exp\Big\{-\frac{1}{2\sigma^2}\sum_i y_i^2+\frac{\beta}{\sigma^2}\sum_i x_iy_i\Big\}.$$

This is a two-parameter exponential family with $S = \big(\sum_i Y_i^2,\ \sum_i x_iY_i\big)$ and natural parameters $\big(\eta_1,\eta_2\big) = \big(-\frac{1}{2\sigma^2},\ \frac{\beta}{\sigma^2}\big)$, whose range is $(-\infty,0)\times\mathbb{R}$ — an open rectangle. By the completeness criterion, S is **complete sufficient**.

The two estimators are functions of S. Minimising $\sum_i(y_i-\beta x_i)^2$ gives

$$\hat\beta = \frac{\sum_i x_iY_i}{\sum_i x_i^2}, \qquad \sum_i(Y_i-\hat\beta x_i)^2 = \sum_i Y_i^2 - \frac{(\sum_i x_iY_i)^2}{\sum_i x_i^2}, \qquad \hat\sigma^2 = \frac{1}{n-1}\sum_i(Y_i-\hat\beta x_i)^2.$$

Both are functions of S alone.

Unbiasedness. $\mathbb{E}\hat\beta = \dfrac{\sum_i x_i\,\beta x_i}{\sum_i x_i^2} = \beta$. For the residual sum of squares, write $\mathbf{Y} = \beta\mathbf{x} + \boldsymbol{\varepsilon}$ and let $P = \mathbf{x}\mathbf{x}^\top/\|\mathbf{x}\|^2$ be the (rank-1) projection onto $\operatorname{span}(\mathbf{x})$. The residual vector is $\mathbf{e} = (I-P)\mathbf{Y} = (I-P)\boldsymbol{\varepsilon}$, so

$$\mathbb{E}\|\mathbf{e}\|^2 = \mathbb{E}\big[\boldsymbol{\varepsilon}^\top(I-P)\boldsymbol{\varepsilon}\big] = \sigma^2\operatorname{tr}(I-P) = \sigma^2(n-1),$$

hence $\mathbb{E}\hat\sigma^2 = \sigma^2$.

By Lehmann–Scheffé, $\hat\beta$ is **the UMVUE of** β and $\hat\sigma^2$ is **the UMVUE of** σ^2. $\square$

Independence: yes. $\hat\beta = \mathbf{a}^\top\mathbf{Y}$ with $\mathbf{a} = \mathbf{x}/\|\mathbf{x}\|^2$, and $\mathbf{e} = (I-P)\mathbf{Y}$. Since $(\hat\beta, \mathbf{e})$ is a linear image of the Gaussian vector $\mathbf{Y}$, it is jointly normal, and

$$\operatorname{Cov}(\hat\beta, \mathbf{e}) = \mathbf{a}^\top\operatorname{Var}(\mathbf{Y})(I-P)^\top = \frac{\sigma^2}{\|\mathbf{x}\|^2}\mathbf{x}^\top(I-P) = \mathbf{0},$$

because $P\mathbf{x} = \mathbf{x}$. Zero covariance plus joint normality gives $\hat\beta \perp\!\!\!\perp \mathbf{e}$, hence $\hat\beta \perp\!\!\!\perp \|\mathbf{e}\|^2 = (n-1)\hat\sigma^2$. Moreover $(n-1)\hat\sigma^2/\sigma^2 \sim \chi^2_{n-1}$, since $I-P$ is a projection of rank $n-1$.

Basu route (worth knowing). Fix σ^2. Then $\hat\beta$ is complete sufficient for β, while $\hat\sigma^2$ has the β-free distribution $\sigma^2\chi^2_{n-1}/(n-1)$ and is therefore ancillary. Basu's theorem gives independence for each fixed σ^2, hence for all (β, σ^2).

Q6. $X_1, \dots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta) = \frac{1}{\theta}e^{-x/\theta}$, $0 < x, \theta < \infty$. Show that $\displaystyle\sum_{i=1}^n X_i$ and $\displaystyle\Big(\sum_{j=1}^p X_j\Big)\Big/\Big(\sum_{k=1}^q X_k\Big)$ are independent for $1 \le p \ne q \le n$.

Key idea

Basu. $\sum X_i$ is complete sufficient; the ratio is scale-invariant and θ is a pure scale parameter, so the ratio is ancillary.

Solution. *Step 1: $S = \sum_{i=1}^n X_i$ is complete sufficient.* The joint density is

$$f(\mathbf{x} \mid \theta) = \theta^{-n}\exp\Big\{-\frac{1}{\theta}\sum_i x_i\Big\}\prod_i \mathbb{I}[x_i > 0],$$

a one-parameter exponential family with $T(x) = x$ and $c(\theta) = -1/\theta$, whose natural parameter space $(-\infty, 0)$ contains an open interval. Hence S is complete sufficient (and $S \sim \text{Gamma}(n, \theta)$).

Step 2: $R = \sum_{j\le p} X_j / \sum_{k\le q} X_k$ is ancillary. θ is a scale parameter: if $Z_1, \dots, Z_n \overset{\text{iid}}{\sim} \text{Exp}(1)$ then $X_i \overset{d}{=} \theta Z_i$ jointly. Substituting,

$$R = \frac{\sum_{j\le p}\theta Z_j}{\sum_{k\le q}\theta Z_k} = \frac{\sum_{j\le p} Z_j}{\sum_{k\le q} Z_k},$$

and the θ's cancel. So the distribution of R does not involve θ: R is ancillary.

Step 3. By Basu's theorem, a complete sufficient statistic is independent of every ancillary statistic. Hence $S \perp\!\!\!\perp R$. $\square$

Watch out

If $p = q$ the ratio is identically 1 and the statement is vacuous, which is why the interesting case is $p \ne q$. Note also that the argument never needs the distribution of R — that is exactly the economy Basu's theorem buys you. (For the record, when the two index sets are disjoint, $\sum_{j\le p} X_j / \sum_k X_k \sim \text{Beta}(p, q)$-type quantities appear; you are not asked for this.)

2 Homework 2 — Minimax, Bayes, and a first MP test

Q1. Prove that for $X \sim N(\theta, 1)$, X is a minimax estimate of θ for $-\infty < \theta < \infty$.

Key idea

X has constant risk 1, but it is not Bayes for any proper prior. So use the *limit of Bayes rules*: take $\pi_k = N(0, k)$, compute the Bayes risk $k/(k+1)$, and let $k \to \infty$.

Solution. *Risk of X.* $R(\theta, X) = \mathbb{E}_\theta(X - \theta)^2 = 1$ for every θ, so $\sup_\theta R(\theta, X) = 1$.

Bayes problem with $\pi_k = N(0, k)$. With $X \mid \theta \sim N(\theta, 1)$ and $\theta \sim N(0, k)$, completing the square in the exponent of $\pi(\theta \mid x) \propto e^{-(x-\theta)^2/2} e^{-\theta^2/2k}$ gives

$$\theta \mid X = x \;\sim\; N\left(\frac{k}{k+1}x,\; \frac{k}{k+1}\right).$$

Hence the Bayes estimate is $T_k(X) = \dfrac{k}{k+1}X$ and, since the Bayes risk under squared error equals the expected posterior variance,

$$r(\pi_k, T_k) = \mathbb{E}_m\big[\operatorname{Var}(\theta \mid X)\big] = \frac{k}{k+1} \qquad \text{(the posterior variance is constant here).}$$

The minimax argument. Let T be *any* estimator. For every k,

$$\sup_{\theta \in \mathbb{R}} R(\theta, T) \;\geq\; \int R(\theta, T)\, \pi_k(\theta)\, d\theta = r(\pi_k, T) \;\geq\; r(\pi_k, T_k) = \frac{k}{k+1},$$

the first inequality because a supremum dominates an average, the second because T_k is Bayes for π_k. Letting $k \to \infty$,

$$\sup_\theta R(\theta, T) \;\geq\; 1 \;=\; \sup_\theta R(\theta, X).$$

Since T was arbitrary, $\inf_T \sup_\theta R(\theta, T) = 1 = \sup_\theta R(\theta, X)$, i.e. X **is minimax**. □

Watch out

X is minimax but *not* admissible in dimension ≥ 3 (Stein). In dimension 1 it is admissible. Also note $\{\pi_k\}$ is called a *least favourable sequence*: no single prior achieves the bound, which is why the sequence version of the theorem is needed here whereas the Binomial problem (Lec 8) can use a single Beta prior.

Q2. Show that for $X \sim \text{Poisson}(\lambda)$, $0 < \lambda < \infty$, any estimate has maximum risk ∞.

Key idea

Same machinery, opposite conclusion: exhibit priors whose Bayes risk diverges. Gamma priors with growing scale do it.

Solution. Take the prior $\pi_{a,b} = \text{Gamma}(a, b)$ (shape a, *scale* b), i.e. $\pi(\lambda) \propto \lambda^{a-1} e^{-\lambda/b}$.

Posterior. $\pi(\lambda \mid x) \propto e^{-\lambda}\lambda^x \cdot \lambda^{a-1} e^{-\lambda/b} = \lambda^{a+x-1} e^{-\lambda(1+1/b)}$, so

$$\lambda \mid X = x \;\sim\; \text{Gamma}\Big(a + x,\; \frac{b}{1+b}\Big),$$

with mean $\frac{b(a+x)}{1+b}$ and variance $\frac{(a+x)b^2}{(1+b)^2}$.

Bayes risk. Since $\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}(X \mid \lambda)\big] = \mathbb{E}[\lambda] = ab$ under the joint law,

$$r(\pi_{a,b}) = \mathbb{E}_m\big[\operatorname{Var}(\lambda \mid X)\big] = \frac{b^2}{(1+b)^2}\,\mathbb{E}[a+X] = \frac{b^2(a+ab)}{(1+b)^2} = \frac{a\,b^2}{1+b}.$$

Divergence. Take $a=1$ and $b=k\to\infty$: $r(\pi_{1,k}) = \dfrac{k^2}{1+k} \to \infty$.
Conclusion. For any estimator T and every k,

$$\sup_{\lambda>0} R(\lambda,T) \;\ge\; r(\pi_{1,k},T) \;\ge\; r(\pi_{1,k}) = \frac{k^2}{1+k} \xrightarrow[k\to\infty]{} \infty.$$

Hence $\sup_{\lambda>0} R(\lambda,T) = \infty$ for every estimator T: the minimax value of the problem is $+\infty$ and no estimator has finite maximum risk. □

Watch out

The intuition: $\operatorname{Var}_\lambda(X) = \lambda \to \infty$ as $\lambda\to\infty$, and the parameter space is unbounded, so no estimator can control the risk uniformly. Contrast with $\operatorname{Bin}(n,\theta)$, where θ lives in the bounded set $(0,1)$ and a genuine minimax estimator exists.

Q3. $X_1,\dots,X_n \overset{\text{iid}}{\sim} \text{Uniform}[\theta-\frac12,\ \theta+\frac12]$. Compute the generalized Bayes estimate under the improper prior $\pi(\theta)=1$, $\theta\in\mathbb{R}$.

Key idea

The likelihood is an indicator. Multiply by the flat prior and read off a uniform posterior on an interval; its mean is the midpoint.

Solution. The likelihood is

$$f(\mathbf{x}\mid\theta) = \prod_{i=1}^n \mathbb{I}\Big[\theta-\tfrac12 \le x_i \le \theta+\tfrac12\Big] = \mathbb{I}\Big[\theta-\tfrac12 \le x_{(1)}\Big]\cdot\mathbb{I}\Big[x_{(n)} \le \theta+\tfrac12\Big] = \mathbb{I}\Big[x_{(n)}-\tfrac12 \;\le\; \theta \;\le\; x_{(1)}+\tfrac12\Big].$$

(The interval is non-empty because the data satisfy $x_{(n)} - x_{(1)} \le 1$ almost surely.)
With $\pi(\theta)=1$,

$$\pi(\theta\mid\mathbf{x}) \;\propto\; f(\mathbf{x}\mid\theta)\cdot 1 = \mathbb{I}\Big[x_{(n)}-\tfrac12 \le \theta \le x_{(1)}+\tfrac12\Big],$$

which is a *proper* density after normalising: the posterior is $\text{Uniform}\big[x_{(n)}-\frac12,\ x_{(1)}+\frac12\big]$, an interval of length $1-(x_{(n)}-x_{(1)})$. The generalized Bayes estimate under squared error is the posterior mean, i.e. the midpoint of this interval:

$$\hat\theta = \frac12\Big[\big(x_{(n)}-\tfrac12\big) + \big(x_{(1)}+\tfrac12\big)\Big] \quad\Longrightarrow\quad \boxed{\hat\theta = \frac{X_{(1)}+X_{(n)}}{2} \quad \text{(the midrange).}}$$

Watch out

Consistent with the general fact from Lecture 7: generalized Bayes estimates are functions of the sufficient statistic, and here $(X_{(1)},X_{(n)})$ is sufficient. Note also that the midrange is *not* the MLE-unique answer — every θ in the interval maximises the likelihood, so the MLE is not unique, while the Bayes approach picks the midpoint canonically.

Q4. Generate $X_1,\dots,X_n \overset{\text{iid}}{\sim} \dfrac{1}{\pi\{1+(x-\theta)^2\}}$ with $\theta=0$, for $n=10,15,20$. Pretending $\theta\in\mathbb{R}$ is unknown, compute the MLE of θ from your generated data.

Key idea

The likelihood equation has no closed form and can have several roots. Start Newton–Raphson from the *sample median* (which is consistent), and confirm the root is the global maximiser by evaluating the log-likelihood on a grid. Never start from $\overline{X}$: for Cauchy data $\overline{X}$ is itself Cauchy(θ) and does not concentrate.

Solution. *The equation.* $\ell(\theta) = -\sum_{i=1}^{n} \log\{1 + (x_i - \theta)^2\} + \text{const}$, so

$$\ell'(\theta) = \sum_{i=1}^{n} \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2} = 0.$$

Clearing denominators gives a polynomial equation of degree $2n - 1$ in θ: no closed form, and up to n local maxima. Newton–Raphson uses

$$\ell''(\theta) = \sum_{i=1}^{n} \frac{2\left[(x_i - \theta)^2 - 1\right]}{\left[1 + (x_i - \theta)^2\right]^2}, \qquad \theta^{(m+1)} = \theta^{(m)} - \frac{\ell'(\theta^{(m)})}{\ell''(\theta^{(m)})}, \qquad \theta^{(0)} = \text{sample median}.$$

Code (Python; an R version is given below).

```
import numpy as np
rng = np.random.default_rng(2026)

def score(t, x): return np.sum(2*(x-t)/(1+(x-t)**2))
def hess (t, x): return np.sum(2*((x-t)**2 - 1)/(1+(x-t)**2)**2)

for n in (10, 15, 20):
    x = rng.standard_cauchy(n) # theta = 0
    t = np.median(x) # consistent starting value
    for _ in range(100):
        step = score(t, x)/hess(t, x)
        t -= step
        if abs(step) < 1e-12: break
    print(n, np.median(x), t, x.mean())
```

```
set.seed(2026)
for (n in c(10,15,20)) {
  x <- rcauchy(n) # theta = 0
  nll <- function(t) sum(log(1 + (x-t)^2))
  fit <- optimize(nll, interval = c(min(x), max(x))) # or optim(median(x), nll)
  cat(n, median(x), fit$minimum, mean(x), "\n")
}
```

Output from one run (numpy generator, seed 2026 — your numbers will differ, the *pattern* is the point):

n	sample median	MLE $\hat{\theta}_n$	sample mean $\overline{X}$	# stationary points
10	−0.4992	−0.4417	−0.544	1
15	−0.3255	0.0316	−0.535	1
20	−0.4850	−0.4267	0.188	1

In every case Newton–Raphson started at the median converged to the global maximiser (verified against a fine grid search over $[\min x_i - 1,\ \max x_i + 1]$).

What to say about the output.

- The MLE hovers near the true $\theta = 0$ but converges slowly — the Cauchy carries little information per observation ($I(\theta) = 1/2$), so the standard error is $\sqrt{2/n} \approx 0.45,\ 0.37,\ 0.32$ for $n = 10, 15, 20$. The observed errors sit comfortably inside one standard error.
- $\overline{X}$ is useless: in the $n = 20$ sample, one observation equal to 20.4 dragged the mean to 0.188 while the bulk of the data sat near -0.5. Indeed $\overline{X} \sim \text{Cauchy}(\theta)$ exactly, for every n — *averaging does not help at all.*
- The sample median is a good, cheap, consistent starting value (Lecture 10); starting Newton–Raphson there avoids converging to a spurious local root.

Q5. Let $X \sim f$ where $H_0 : f(x) = \frac{1}{2}e^{-|x|}$ (Laplace) and $H_A : f(x) = N(0, 1)$. Derive the most powerful test of level $0 < \alpha < 1$.

Key idea

Form the likelihood ratio and simplify: it is a function of $|x|$ that *is not monotone* — it rises to a peak at $|x| = 1$ and falls away. So the rejection region is an *annulus* $a < |X| < b$, not a tail. Use $\mathbb{P}_0(|X| > t) = e^{-t}$ to fix the constants.

Solution. *Step 1: the likelihood ratio.*

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)} = \frac{(2\pi)^{-1/2}e^{-x^2/2}}{\frac{1}{2}e^{-|x|}} = \sqrt{\frac{2}{\pi}}\ \exp\left\{|x| - \frac{x^2}{2}\right\}.$$

So Λ is an increasing function of $g(u) := u - \frac{u^2}{2}$ evaluated at $u = |x| \geq 0$.

Step 2: shape of the rejection region. $g'(u) = 1 - u$, so g increases on $[0, 1]$, peaks at $g(1) = \frac{1}{2}$, and decreases thereafter, with $g(u) \to -\infty$. Hence $\{g(u) > c\}$ is an *interval* in u:

$$\{u \geq 0 :\ g(u) > c\} = \{u :\ 1 - \sqrt{1 - 2c} < u < 1 + \sqrt{1 - 2c}\} \cap [0, \infty), \qquad c < \tfrac{1}{2}.$$

Writing $a = 1 - \sqrt{1 - 2c}$ and $b = 1 + \sqrt{1 - 2c}$, note $a + b = 2$, so the two cutoffs satisfy $b = 2 - a$. The MP test therefore rejects H_0 when

$$a < |X| < b = 2 - a.$$

Both distributions are continuous, so no randomisation is needed.

Step 3: fix the size. Under H_0, $\mathbb{P}_0(|X| > t) = e^{-t}$ for $t \geq 0$. Hence for $0 \leq a < b$,

$$\mathbb{P}_0\big(a < |X| < b\big) = e^{-a} - e^{-b}.$$

Case (i): $a \geq 0$, i.e. $c \geq 0$. Set $h(a) := e^{-a} - e^{-(2-a)} = e^{-a} - e^{a-2}$ for $a \in [0, 1]$. Then h is strictly decreasing with $h(0) = 1 - e^{-2} \approx 0.8647$ and $h(1) = 0$. So for every $\alpha \in \big(0, 1 - e^{-2}\big)$ there is a unique $a = a(\alpha) \in (0, 1)$ solving $e^{-a} - e^{a-2} = \alpha$, and the MP test is

$$\boxed{\phi^*(x) = \mathbb{I}\big[a < |x| < 2 - a\big], \qquad e^{-a} - e^{a-2} = \alpha.}$$

Case (ii): $\alpha \geq 1 - e^{-2}$. Then $c < 0$, the constraint $|x| > a$ becomes vacuous, and the region is $\{|X| < b\}$ with $\mathbb{P}_0(|X| < b) = 1 - e^{-b} = \alpha$, i.e. $b = -\log(1 - \alpha) > 2$.

Step 4: power. Under H_A, $X \sim N(0, 1)$, so in Case (i)

$$\beta = \mathbb{P}_1\big(a < |X| < 2 - a\big) = 2\big[\Phi(2 - a) - \Phi(a)\big].$$

Numerical illustration. For $\alpha = 0.05$, solving $e^{-a} - e^{a-2} = 0.05$ gives $a \approx 0.9321$, so the test rejects when $0.932 < |X| < 1.068$, with power $2[\Phi(1.068) - \Phi(0.932)] \approx 0.0657$. The power barely exceeds the size — a Laplace and a standard normal are genuinely hard to tell apart from one observation, which is exactly the lesson the problem is designed to deliver.

Watch out

The instinctive answer "reject in the tails" is **wrong here**. The normal has *lighter* tails than the Laplace, so large $|x|$ is evidence *for* H_0; the normal is relatively more likely at moderate $|x|$ near 1. Always sketch Λ before declaring the shape of the region.

3 Homework 3 — MP and UMP tests

Q1. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \frac{1}{\theta} e^{-x/\theta}$, $0 < x, \theta < \infty$. Derive the form of the UMP test of $H_0 : \theta \le 1$ against $H_A : \theta > 1$ with given size $0 < \alpha < 1$.

Solution. *Step 1: MLR.* The joint density is $f(\mathbf{x} \mid \theta) = \theta^{-n} \exp\{-\frac{1}{\theta} \sum x_i\}$, a one-parameter exponential family with $T(\mathbf{x}) = \sum_i x_i$ and $c(\theta) = -1/\theta$, which is *strictly increasing* in θ. Explicitly, for $\theta_1 > \theta_0$,

$$\frac{f(\mathbf{x} \mid \theta_1)}{f(\mathbf{x} \mid \theta_0)} = \Big(\frac{\theta_0}{\theta_1}\Big)^n \exp\Big\{\Big(\frac{1}{\theta_0} - \frac{1}{\theta_1}\Big) \sum_i x_i\Big\},$$

which is increasing in $\sum_i x_i$ because $\frac{1}{\theta_0} - \frac{1}{\theta_1} > 0$. So the family has **MLR in** $T = \sum_i X_i$.

Step 2: Karlin–Rubin. The UMP level-α test of $H_0 : \theta \le 1$ vs $H_A : \theta > 1$ is

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & T(\mathbf{x}) > c, \\ 0, & T(\mathbf{x}) \le c, \end{cases} \qquad \text{with } \mathbb{P}_{\theta=1}(T > c) = \alpha.$$

No randomisation is needed: T is continuous.

Step 3: the cutoff. Under θ, $T = \sum_i X_i \sim \text{Gamma}(n, \theta)$ and $\dfrac{2T}{\theta} \sim \chi^2_{2n}$. At the boundary $\theta = 1$ this gives $2T \sim \chi^2_{2n}$, so

$$\mathbb{P}_{\theta=1}(T > c) = \mathbb{P}\big(\chi^2_{2n} > 2c\big) = \alpha \quad \Longrightarrow \quad c = \tfrac{1}{2}\chi^2_{2n,\alpha},$$

where $\chi^2_{2n,\alpha}$ is the upper-α point. Hence

$$\boxed{\text{Reject } H_0 \iff 2\sum_{i=1}^n X_i > \chi^2_{2n,\alpha} \quad \text{i.e.} \quad \sum_i X_i > \tfrac{1}{2}\chi^2_{2n,\alpha}.}$$

Step 4: size and power. The power function is $\beta(\theta) = \mathbb{P}\big(\chi^2_{2n} > \chi^2_{2n,\alpha}/\theta\big)$, which is increasing in θ. Hence $\sup_{\theta \le 1} \beta(\theta) = \beta(1) = \alpha$: the test has size exactly α over the composite null, and $\beta(\theta) \to 1$ as $\theta \to \infty$.

Q2. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \dfrac{\lambda^x e^{-\lambda}}{x!}$, $\lambda > 0$. Derive the form of the UMP test of $H_0 : \lambda \le 2$ against $H_A : \lambda > 2$ with given size $0 < \alpha < 1$.

Solution. *Step 1: MLR.* The joint pmf is

$$f(\mathbf{x} \mid \lambda) = e^{-n\lambda} \, \frac{1}{\prod_i x_i!} \, \exp\Big\{\log\lambda \sum_i x_i\Big\},$$

a one-parameter exponential family with $T = \sum_i X_i$ and $c(\lambda) = \log\lambda$, strictly increasing. So the family has **MLR in** $T = \sum_i X_i$, and $T \sim \text{Poisson}(n\lambda)$.

Step 2: Karlin–Rubin, with randomisation. The UMP level-α test is

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & T > c, \\ \gamma, & T = c, \\ 0, & T < c, \end{cases} \qquad \text{with} \quad \mathbb{P}_{\lambda=2}(T > c) + \gamma\, \mathbb{P}_{\lambda=2}(T = c) = \alpha.$$

Under $\lambda = 2$ we have $T \sim \text{Poisson}(2n)$, so writing $P(t) = \mathbb{P}\big(\text{Poi}(2n) > t\big)$ and $p(t) = \mathbb{P}\big(\text{Poi}(2n) = t\big)$:

$$\boxed{c = \min\{t \in \mathbb{Z}_{\ge 0} : \ P(t) \le \alpha\}, \qquad \gamma = \frac{\alpha - P(c)}{p(c)} \in [0, 1).}$$

Step 3: why randomisation is unavoidable. T is discrete, so $\mathbb{P}_{\lambda=2}(T > c)$ takes only countably many values and will not generally equal α. Randomising on the boundary point $T = c$ restores exact size α; without it the test is merely of level α and is no longer UMP among all level-α tests. (In practice one reports the non-randomised version, rejecting iff $T > c$, at the conservative level $P(c) \le \alpha$.)

Step 4: size over the composite null. The power function $\beta(\lambda) = \mathbb{E}_\lambda \phi^*$ is non-decreasing in λ (MLR), so $\sup_{\lambda \le 2} \beta(\lambda) = \beta(2) = \alpha$.

Worked instance ($n = 5$, $\alpha = 0.05$). Then $T \sim \text{Poi}(10)$ under H_0; $\mathbb{P}(T > 15) = 0.0487$ and $\mathbb{P}(T > 14) = 0.0835$, so $c = 15$, $p(15) = \mathbb{P}(T = 15) = 0.0347$, and $\gamma = (0.05 - 0.0487)/0.0347 \approx 0.036$. Reject if $T \ge 16$; if $T = 15$, reject with probability 0.036.

Q3. $H_0 : X \sim p_0(x) = \left(\frac{1}{2}\right)^{x+1}$, $x = 0, 1, 2, \ldots$ against $H_A : X \sim p_1(x) = \dfrac{e^{-1}}{x!}$, $x = 0, 1, 2, \ldots$. Derive the form of the MP test of given size $0 < \alpha < 1$.

Key idea

Two simple hypotheses $\Rightarrow$ pure Neyman–Pearson. Compute $\Lambda(x) = p_1/p_0$, sort the sample space in *decreasing* order of Λ, and fill the rejection region greedily until the null probability reaches α. The ratio here is not monotone in x, and it has a *tie*.

Solution. *Step 1: the likelihood ratio.*

$$\Lambda(x) = \frac{p_1(x)}{p_0(x)} = \frac{e^{-1}/x!}{2^{-(x+1)}} = \frac{2}{e} \cdot \frac{2^x}{x!}.$$

So Λ is an increasing function of $\lambda(x) := 2^x/x!$. Tabulate:

x	0	1	2	3	4	5	6	≥ 7
$\lambda(x) = 2^x/x!$	1	2	2	1.333	0.667	0.267	0.089	$\downarrow 0$
$p_0(x) = 2^{-(x+1)}$	1/2	1/4	1/8	1/16	1/32	1/64	1/128	

Ordering the sample space by *decreasing* λ:

$$\{1,2\}\ (\lambda = 2)\ \succ\ \{3\}\ (1.333)\ \succ\ \{0\}\ (1)\ \succ\ \{4\}\ (0.667)\ \succ\ \{5\}\ \succ\ \{6\} \succ \cdots$$

Note the **tie** at $x = 1$ and $x = 2$, and note that $x = 0$ sits *below* $x = 3$ — the region is not an upper tail.

Step 2: cumulative null probabilities along this ordering.

$$\mathbb{P}_0\{1,2\} = \tfrac{1}{4} + \tfrac{1}{8} = \tfrac{3}{8}, \quad +\mathbb{P}_0\{3\} = \tfrac{3}{8} + \tfrac{1}{16} = \tfrac{7}{16}, \quad +\mathbb{P}_0\{0\} = \tfrac{7}{16} + \tfrac{1}{2} = \tfrac{15}{16}, \quad +\mathbb{P}_0\{4\} = \tfrac{31}{32}, \ \ldots$$

Step 3: the test. The MP test has the Neyman–Pearson form $\phi^*(x) = 1, \gamma, 0$ according as $\Lambda(x) > k, = k, < k$, with k, γ chosen so that $\mathbb{E}_0 \phi^* = \alpha$. Explicitly:

$$\boxed{\begin{array}{ll}
0 < \alpha \le \frac{3}{8}: & \phi^* = \frac{8\alpha}{3} \text{ on } \{1,2\}, \quad 0 \text{ elsewhere} \\
\frac{3}{8} < \alpha \le \frac{7}{16}: & \phi^* = 1 \text{ on } \{1,2\}, \ \ \phi^* = 16\alpha - 6 \text{ on } \{3\}, \ 0 \text{ else} \\
\frac{7}{16} < \alpha \le \frac{15}{16}: & \phi^* = 1 \text{ on } \{1,2,3\}, \ \ \phi^* = 2\alpha - \frac{7}{8} \text{ on } \{0\}, \ 0 \text{ else} \\
\frac{15}{16} < \alpha \le \frac{31}{32}: & \phi^* = 1 \text{ on } \{0,1,2,3\}, \ \ \phi^* = 32\alpha - 30 \text{ on } \{4\}, \ 0 \text{ else}
\end{array}}$$

and so on. (Each randomisation constant is $\big(\alpha - \mathbb{P}_0[\text{region so far}]\big)/p_0[\text{boundary point}]$.)

Power. For instance at $\alpha = 0.05$: $\phi^* = 8(0.05)/3 = 0.1333$ on $\{1,2\}$, so the power is $0.1333 \times \big[p_1(1) + p_1(2)\big] = 0.1333 \times \big[e^{-1} + \frac{e^{-1}}{2}\big] = 0.1333 \times 0.5518 = 0.0736$.

Watch out

Two things to notice.
(a) The region is not a tail. $x = 0$ is ranked *below* $x = 3$, so for α between $7/16$ and $15/16$ the rejection region is $\{0, 1, 2, 3\}$ — it includes both ends and excludes nothing in between only by accident. Never assume the NP region is one-sided in x; it is one-sided in Λ.
(b) The tie is where NP is indifferent. Since $\Lambda(1) = \Lambda(2) = 4/e$, we have $p_1(x) = \frac{4}{e}p_0(x)$ on the tie set, so *any* allocation (γ_1, γ_2) with $\gamma_1 p_0(1) + \gamma_2 p_0(2) = \alpha$ gives the same power $\frac{4}{e}\big[\gamma_1 p_0(1) + \gamma_2 p_0(2)\big] = \frac{4}{e}\alpha$. All such tests are MP; the uniform choice $\gamma_1 = \gamma_2 = 8\alpha/3$ is just the tidy convention. (Check at $\alpha = 0.05$: $\frac{4}{e} \times 0.05 = 0.0736$, matching the power computed above.)

Q4. $X \sim \text{Uniform}[0, \theta]$. Derive the form of the UMP test of $H_0 : \theta = 1$ against $H_A : \theta \neq 1$ with given size $0 < \alpha < 1$.

Key idea

The support moves with θ. Rejecting on $\{X > 1\}$ costs *nothing* under H_0 and is forced by NP against every $\theta > 1$; for $\theta < 1$ the likelihood ratio is flat on $[0, \theta]$ so the best test puts its α as far left as possible. The two requirements are compatible, and a two-sided UMP test exists — the standard exception to the usual rule.

Solution. Write $f_\theta(x) = \frac{1}{\theta}\mathbb{I}[0 \le x \le \theta]$, so $f_1(x) = \mathbb{I}[0 \le x \le 1]$. Consider the test

$$\boxed{\phi^*(x) = \begin{cases} 1, & x \le \alpha \quad \text{or} \quad x > 1, \\ 0, & \alpha < x \le 1. \end{cases}}$$

Its size is $\mathbb{P}_{\theta=1}(X \le \alpha) + \mathbb{P}_{\theta=1}(X > 1) = \alpha + 0 = \alpha$. ✓

Optimality against $\theta_1 > 1$. Here

$$\frac{f_{\theta_1}(x)}{f_1(x)} = \begin{cases} 1/\theta_1, & 0 \le x \le 1, \\ +\infty, & 1 < x \le \theta_1. \end{cases}$$

By Neyman–Pearson the MP test must reject wherever the ratio is $+\infty$, i.e. on $(1, \theta_1]$, and is otherwise *indifferent* on $[0, 1]$ because the ratio is the constant $1/\theta_1$ there. So *any* test with $\phi = 1$ on $\{x > 1\}$ and $\int_0^1 \phi(x)\,dx = \alpha$ is MP of size α; ϕ^* is one such. Its power is

$$\beta(\theta_1) = \frac{\theta_1 - 1}{\theta_1} + \frac{\alpha}{\theta_1} = 1 - \frac{1-\alpha}{\theta_1},$$

which is the maximum achievable. This holds for every $\theta_1 > 1$ simultaneously.

Optimality against $\theta_1 < 1$. Here

$$\frac{f_{\theta_1}(x)}{f_1(x)} = \begin{cases} 1/\theta_1, & 0 \le x \le \theta_1, \\ 0, & \theta_1 < x \le 1, \end{cases}$$

so the MP test rejects for small x: it must place all its rejection mass inside $[0, \theta_1]$ as far as the size permits. For any level-α test ϕ (so $\int_0^1 \phi = \alpha$) the power is

$$\mathbb{E}_{\theta_1}\phi = \frac{1}{\theta_1}\int_0^{\theta_1} \phi(x)\,dx \;\le\; \frac{\min(\alpha, \theta_1)}{\theta_1},$$

since $\int_0^{\theta_1} \phi \le \min\left(\int_0^1 \phi,\ \theta_1\right) = \min(\alpha, \theta_1)$. Our ϕ^* achieves this bound: if $\theta_1 \ge \alpha$ then $\int_0^{\theta_1} \phi^* = \alpha$ and the power is α/θ_1; if $\theta_1 < \alpha$ then $\phi^* \equiv 1$ on $[0, \theta_1]$ and the power is 1. Again this holds for every $\theta_1 < 1$ simultaneously.

Conclusion. ϕ^* is MP against every $\theta \ne 1$ at level α, hence it is **UMP of size** α for $H_0 : \theta = 1$ vs $H_A : \theta \ne 1$. Its power function is

$$\beta(\theta) = \begin{cases} 1, & \theta \le \alpha, \\ \alpha/\theta, & \alpha < \theta \le 1, \\ 1 - (1-\alpha)/\theta, & \theta > 1, \end{cases}$$

with $\beta(1) = \alpha$. □

Watch out

This is the standard example showing that "no UMP test exists for two-sided alternatives" is a statement about *regular* (MLR) families, not a theorem of nature. The mechanism is that the support depends on θ: the event $\{X > 1\}$ is impossible under H_0, so including it in the rejection region is free, and it is precisely what the alternatives $\theta > 1$ reward. The piece of the region inside $[0, 1]$ is then free to be placed where the $\theta < 1$ alternatives want it. Note ϕ^* is *not* unique: on $[0, 1]$ any allocation of α is MP against $\theta > 1$, but only the left-most allocation $[0, \alpha]$ also works against $\theta < 1$. That is what pins the test down.

Companion document: `PI_Exam_Revision.pdf` *— theory, proofs and cheat sheets.*